

MAH, Commercialization Frictions, and Original-Drug Innovation

Abstract

This note develops a parsimonious model of how China's Marketing Authorization Holder (MAH) system can affect original-drug innovation. The model keeps the institutional timing explicit. Firms choose original-drug R&D intensity before successful drug opportunities are realized. If a successful opportunity is realized, the firm chooses a commercialization route: internal production, entrusted production while retaining marketing authorization, transfer/licensing/sale, or abandonment. MAH enters the model in two linked ways. First, it makes the MAH-style entrusted-production route part of the relevant commercialization choice set. Second, conditional on this route being institutionally available, MAH lowers the commercialization friction of using a qualified entrusted manufacturer while retaining authorization. The first component captures route-set expansion; the second component yields derivative-based comparative statics with respect to the entrusted-production friction. The reform does not directly raise scientific productivity, lower clinical-trial cost, or make entrusted production universally better than internal production or transfer. Instead, it improves one option in the firm's expected commercialization problem. Because the firm chooses the maximum over routes, the expected value of a successful original-drug opportunity weakly rises. This higher expected future commercialization value enters the Bellman equation and increases current R&D intensity. The only endogenous state is the stock of retained original drugs or certificates. Research ability and manufacturing ability are fixed firm characteristics drawn from an arbitrary joint distribution. The model yields closed-form solutions for R&D intensity, expected original-drug opportunities, incumbent value, entry value, aggregate drug counts, and derivative-based comparative statics. It also states a modest calibration strategy and possible datasets without claiming full structural identification.

Keywords: MAH; original-drug innovation; commercialization frictions; entrusted production; route choice; firm entry.

JEL codes: D22; L65; O31; O38; I18.

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1 Introduction

The purpose of this theory section is to discipline a model-guided empirical design rather than to estimate a full dynamic structural model of the Chinese pharmaceutical industry. The central question is whether MAH raised original-drug innovation by improving downstream commercialization options. The mechanism is an expected-profit channel. A firm chooses R&D before knowing how many drug opportunities will be successful, but it anticipates the institutional regime under which a successful opportunity will later be produced, entrusted, transferred, or abandoned.

This logic follows two lines of literature. First, pharmaceutical innovation responds to expected downstream rewards. Market-size and expected-return models show that firms direct innovation toward projects with higher expected commercial value ([Schmookler, 1966](#); [Acemoglu and Linn, 2004](#); [Finkelstein, 2004](#); [Blume-Kohout and Sood, 2013](#); [Dubois et al., 2015](#); [Budish et al., 2015](#); [Dranove et al., 2022](#)). Second, the value of innovation depends on complementary assets and commercialization strategy. An innovator without manufacturing or distribution capability may need to cooperate with, license to, or sell to firms that possess those assets ([Teece, 1986](#); [Arora et al., 2001](#); [Gans and Stern, 2003](#)). MAH fits this logic because it changes the commercialization environment by allowing the holder to retain authorization while using a qualified external manufacturer.

The model is deliberately narrow. It does not treat MAH as a demand shock. It does not assume that MAH makes science easier. It does not assume that entrusted production dominates internal production. It also does not track every product in a firm’s portfolio. Instead, it follows the product-line innovation tradition by using a stock of products or certificates as a low-dimensional state and an innovation-arrival function linking R&D effort to expected new drug opportunities ([Klette and Kortum, 2004](#)). The resulting framework is suitable for a setting where the data contain certificate counts, approval records, holder-producer links, firm characteristics, and entry, but not project-level R&D expenditure or project-specific success probabilities.

2 Related Literature

The first relevant literature studies how expected rewards shape pharmaceutical innovation. [Acemoglu and Linn \(2004\)](#) show that potential market size affects entry of new drugs, and later work studies how insurance design, reimbursement, and expected profitability shift drug-development incentives ([Finkelstein, 2004](#); [Blume-Kohout and Sood, 2013](#); [Dubois et al., 2015](#); [Budish et al., 2015](#); [Dranove et al., 2022](#)). Recent work on China studies how regulatory and demand-side changes affect drug development ([Jia et al., 2023](#); [Barwick et al., 2026](#)). This paper differs by focusing on a commercialization-friction channel rather than on market size, approval delay, or reimbursement.

The second literature studies commercialization strategy and complementary assets. Teece (1986) emphasizes that innovators' returns depend on access to complementary assets. Arora et al. (2001) and Gans and Stern (2003) analyze markets for technology and the choice between commercialization and cooperation with established firms. The MAH reform is naturally interpreted through this lens: it improves one way for an innovator to access manufacturing capability while retaining the authorization asset.

The third literature provides the modeling style. Klette and Kortum (2004) use stochastic innovation arrivals and product-line growth to connect firm-level R&D to aggregate innovation. The model below uses a simpler version of that idea: R&D intensity determines the expected number of original-drug opportunities, and retained opportunities enter a drug-stock state. The model avoids the full state space of dynamic industry models such as Ericson and Pakes (1995) and Bajari et al. (2007).

3 Institutional Background and Economic Timing

The model distinguishes legal registration timing from economic decision timing. Under China's *Provisions for Drug Registration*, drug registration includes applications for clinical trials, marketing authorization, renewal, and supplementary applications. After obtaining the Drug Approval License, the applicant becomes the MAH (National Medical Products Administration, 2022a). Therefore, the model should not say that a firm must already hold the final approval or a commercial manufacturing certificate before it starts R&D. The relevant point is forward-looking: expected future commercialization feasibility affects the value of current R&D.

Under the *Drug Administration Law*, an MAH may manufacture drug products by itself or entrust a qualified drug manufacturer. If production is contracted out, the MAH and the contract manufacturer must sign entrustment and quality agreements, and the MAH remains responsible for the drug's safety, efficacy, and quality (National Medical Products Administration, 2019). The manufacturing supervision rules require the MAH to assess the contracted firm's quality assurance and risk-management capabilities, sign quality and entrustment agreements, and supervise performance (National Medical Products Administration, 2022b). Thus MAH does not remove regulation. It creates a regulated holder-producer separation route.

The economic timing is as follows. In period t , a firm chooses original-drug R&D intensity. In period $t + 1$, successful original-drug opportunities may arrive. Conditional on an opportunity, the firm chooses among internal production, entrusted production while retaining MAH, transfer/licensing/sale, or abandonment. This timing captures the expectation mechanism: MAH facilitates expected future commercialization profits, and this expected future value motivates more current R&D.

4 Model

4.1 Firm characteristics and state variables

Each firm i has fixed research and manufacturing abilities,

$$(a_i, k_i) \sim H(a, k), \quad (1)$$

where $a_i > 0$ is research ability and $k_i > 0$ is manufacturing ability. The joint distribution H is unrestricted. Research ability and manufacturing ability may be positively correlated, negatively correlated, or independent. This is important because the model should not impose that research-oriented firms always lack manufacturing capability.

The endogenous state is

$$n_{it} \geq 0, \quad (2)$$

where n_{it} is the stock of retained original drugs or original-drug certificates held by firm i at the beginning of period t . Existing retained drugs generate flow payoff

$$\pi n_{it}, \quad \pi > 0. \quad (3)$$

The stock depreciates, expires, or becomes obsolete at rate $\delta \in (0, 1)$. This product-stock state is a low-dimensional substitute for a full multi-product portfolio, following the spirit of product-line innovation models ([Klette and Kortum, 2004](#)).

4.2 R&D production function and uncertainty

The firm chooses original-drug R&D intensity

$$x_{it} \geq 0. \quad (4)$$

R&D intensity produces a random number of successful original-drug opportunities in the next period. Let $Y_{i,t+1}$ denote the number of such opportunities. The drug-count production function is

$$\lambda_{i,t+1}(x_{it}) \equiv \mathbb{E}_t[Y_{i,t+1}] = f(a_i, x_{it}) = a_i x_{it}. \quad (5)$$

Equation (5) is the model's production function for the number of new original-drug opportunities or certificates. It is not physical pill production. It says that research effort raises the expected number of successful opportunities, and higher research ability makes effort more productive. This is the simplest version of the stochastic innovation-arrival structure used in product-line innovation models ([Klette and Kortum, 2004](#)). A

Poisson count interpretation is optional,

$$Y_{i,t+1} \mid x_{it}, a_i \sim \text{Poisson}(a_i x_{it}), \quad (6)$$

but the comparative statics below require only the conditional mean in equation (5). Any unobserved scientific success probability can be absorbed into the effective research ability a_i .

R&D effort has convex cost,

$$C(x_{it}) = \frac{\kappa}{2} x_{it}^2, \quad \kappa > 0. \quad (7)$$

This reduced-form cost delivers an interior closed-form solution. It is not a claim that project-level R&D spending is observed. Reduced-form innovation-effort costs are common in expected-return models of innovation ([Acemoglu and Linn, 2004](#); [Klette and Kortum, 2004](#); [Jia et al., 2023](#)).

4.3 MAH indicator, route availability, and entrusted-production friction

Let $M_{t+1} \in \{0, 1\}$ denote the institutional regime relevant for commercialization in period $t + 1$. The variable $M_{t+1} = 0$ denotes the pre-MAH regime, and $M_{t+1} = 1$ denotes the MAH regime. The MAH-style entrusted-production route is available only when $M_{t+1} = 1$. The feasible route set is

$$\mathcal{R}(M_{t+1}) = \{I, T, A\} \cup \{E : M_{t+1} = 1\}, \quad (8)$$

where I is internal production, E is entrusted production while retaining MAH, T is transfer/licensing/sale, and A is abandonment or delay.

Conditional on the entrusted route being available, MAH also lowers the friction of using that route. A convenient way to write this is

$$\tau_{i,t+1}^E(M_{t+1}) = \bar{\tau}_i^E - \Delta\tau_i M_{t+1}, \quad \Delta\tau_i \geq 0. \quad (9)$$

Equation (9) should be read together with the route set in equation (8). When $M_{t+1} = 0$, the MAH-style entrusted route is not feasible for the innovating holder. When $M_{t+1} = 1$, the route is feasible and its friction is lower. This formulation captures both ideas: MAH adds a regulated holder-producer separation option, and the strength of the option is governed by the friction τ^E .

For derivative comparative statics, it is useful to study Γ as a function of the continuous friction τ^E conditional on E being feasible. The binary reform effect can then be

decomposed into route-set expansion plus a fall in τ^E .

4.4 Commercialization route choice in period $t + 1$

For each successful original-drug opportunity, the firm chooses one route $r \in \mathcal{R}(M_{t+1})$. Let R_{t+1} be the gross expected market return from a commercialized original drug in period $t + 1$. This object summarizes market size, therapeutic value, reimbursement environment, and product demand. It is included because pharmaceutical innovation responds to expected market rewards (Acemoglu and Linn, 2004; Dubois et al., 2015; Barwick et al., 2026). The model does not require firm-product revenue data.

Current route payoffs in period $t + 1$ are

$$b_{i,t+1}^I = R_{t+1} - C^I(k_i), \quad (10)$$

$$b_{i,t+1}^E = R_{t+1} - C^E - \tau_{i,t+1}^E(M_{t+1}), \quad (11)$$

$$b_{i,t+1}^T = S_i - \tau_i^T, \quad (12)$$

$$b_{i,t+1}^A = 0. \quad (13)$$

Here $C^I(k_i)$ is internal production cost, with

$$\frac{\partial C^I(k_i)}{\partial k_i} < 0, \quad (14)$$

so higher manufacturing ability lowers internal production cost. C^E is the direct cost of entrusted production, and $\tau_{i,t+1}^E$ is the institutional friction of retaining authorization while using a qualified entrusted manufacturer. $S_i - \tau_i^T$ is the net value of transfer, sale, or licensing.

If the firm chooses internal production or entrusted production, it retains the drug in its own future stock. If it transfers, sells, licenses, or abandons, the innovating firm does not retain it. Define

$$\rho^I = \rho^E = 1, \quad \rho^T = \rho^A = 0. \quad (15)$$

Given the marginal value v_{t+1} of one retained original drug at the beginning of period $t + 1$, the route-specific value of a successful opportunity is

$$G_{i,t+1}^r = b_{i,t+1}^r + v_{t+1}\rho^r. \quad (16)$$

The firm chooses

$$r_{i,t+1}^* \in \arg \max_{r \in \mathcal{R}(M_{t+1})} G_{i,t+1}^r. \quad (17)$$

Define the expected value of one successful original-drug opportunity as

$$\Gamma_{i,t+1}(M_{t+1}, \tau_{i,t+1}^E) = \max_{r \in \mathcal{R}(M_{t+1})} \{b_{i,t+1}^r + v_{t+1}\rho^r\}. \quad (18)$$

When $M_{t+1} = 1$, this becomes

$$\Gamma_{i,t+1}^1(\tau_{i,t+1}^E) = \max \{R_{t+1} - C^I(k_i) + v_{t+1}, R_{t+1} - C^E - \tau_{i,t+1}^E + v_{t+1}, S_i - \tau_i^T, 0\}. \quad (19)$$

When $M_{t+1} = 0$, the MAH-style entrusted route is not available:

$$\Gamma_{i,t+1}^0 = \max \{R_{t+1} - C^I(k_i) + v_{t+1}, S_i - \tau_i^T, 0\}. \quad (20)$$

Equations (19) and (20) show both parts of the mechanism. MAH adds the entrusted-production route to the commercialization choice set. Conditional on that route being available, a lower τ^E increases its value. The model does not imply that entrusted production is always chosen.

4.5 Recursive Bellman equation

The state transition is

$$n_{i,t+1} = (1 - \delta)n_{it} + \rho_{i,t+1}^* Y_{i,t+1}, \quad (21)$$

where $\rho_{i,t+1}^* = \rho^{r_{i,t+1}^*}$. Equation (21) is written in expected-count form. If one wants to track multiple realized opportunities one by one, the last term can be written as a summation over realized opportunities. That notation is not needed for the main model.

The recursive problem is

$$\begin{aligned} V_t(n_{it}; a_i, k_i, S_i, \tau_i^T, M_t) = \max_{x_{it} \geq 0} & \left\{ \pi n_{it} - \frac{\kappa}{2} x_{it}^2 \right. \\ & \left. + \beta \mathbb{E}_t \left[V_{t+1}(n_{i,t+1}; a_i, k_i, S_i, \tau_i^T, M_{t+1}) + Y_{i,t+1} b_{i,t+1}^{r_{i,t+1}^*} \right] \right\}. \end{aligned} \quad (22)$$

This equation shows the mechanism directly. Current R&D effort x_{it} raises the expected number of next-period successful opportunities. Those opportunities are valued through the next-period route decision, which is affected by MAH through both $\mathcal{R}(M_{t+1})$ and $\tau_{i,t+1}^E(M_{t+1})$. Thus expected future commercialization profits enter the current R&D choice.

Because the value function is affine in the stock state, write

$$V_t(n_{it}; \cdot) = A_t(\cdot) + v_t n_{it}. \quad (23)$$

The marginal value of one retained original drug satisfies

$$v_t = \pi + \beta(1 - \delta)v_{t+1}. \quad (24)$$

Using equations (5), (18), and (24), the Bellman reduces to

$$\begin{aligned} V_t(n_{it}; \cdot) = & \pi n_{it} + \beta A_{t+1} + \beta(1 - \delta)v_{t+1}n_{it} \\ & + \max_{x_{it} \geq 0} \left\{ \beta a_i x_{it} \mathbb{E}_t [\Gamma_{i,t+1}(M_{t+1}, \tau_{i,t+1}^E)] - \frac{\kappa}{2} x_{it}^2 \right\}. \end{aligned} \quad (25)$$

The reduced problem keeps the same economics as the full Bellman but makes the expected-profit channel transparent.

5 Closed-Form Solution

The R&D choice solves

$$\max_{x_{it} \geq 0} \left\{ \beta a_i x_{it} \mathbb{E}_t [\Gamma_{i,t+1}] - \frac{\kappa}{2} x_{it}^2 \right\}. \quad (26)$$

The first-order condition is

$$\beta a_i \mathbb{E}_t [\Gamma_{i,t+1}] - \kappa x_{it} = 0. \quad (27)$$

Since abandonment gives value zero, $\Gamma_{i,t+1} \geq 0$. The optimal R&D intensity is

$$x_{it}^* = \frac{\beta a_i}{\kappa} \mathbb{E}_t [\Gamma_{i,t+1}]. \quad (28)$$

The expected number of next-period successful original-drug opportunities is

$$\lambda_{i,t+1}^* \equiv \mathbb{E}_t [Y_{i,t+1}^*] = a_i x_{it}^* = \frac{\beta a_i^2}{\kappa} \mathbb{E}_t [\Gamma_{i,t+1}]. \quad (29)$$

Equations (28) and (29) give the mechanism in closed form: expected future commercialization value raises current R&D and expected future original-drug counts.

For stationary comparative statics, set $R_t = R$, $M_t = M$, $\tau_{it}^E = \tau^E$, $v_t = v$, and $A_t = A$. From equation (24),

$$v = \frac{\pi}{1 - \beta(1 - \delta)}. \quad (30)$$

The post-MAH opportunity value as a function of the continuous entrusted-production friction is

$$\Gamma_i^1(k_i, \tau_i^E) = \max \left\{ R - C^I(k_i) + v, R - C^E - \tau_i^E + v, S_i - \tau_i^T, 0 \right\}. \quad (31)$$

The pre-MAH opportunity value is

$$\Gamma_i^0(k_i) = \max \left\{ R - C^I(k_i) + v, S_i - \tau_i^T, 0 \right\}. \quad (32)$$

The stationary solutions under regime M are

$$x_i^*(M) = \frac{\beta a_i}{\kappa} \Gamma_i^M, \quad (33)$$

$$\lambda_i^*(M) = \frac{\beta a_i^2}{\kappa} \Gamma_i^M, \quad (34)$$

$$V_i(n_i; M) = \frac{\beta^2 a_i^2}{2\kappa(1-\beta)} \left[\Gamma_i^M \right]^2 + \frac{\pi}{1-\beta(1-\delta)} n_i. \quad (35)$$

where $\Gamma_i^M = \Gamma_i^0$ if $M = 0$ and $\Gamma_i^M = \Gamma_i^1(k_i, \tau_i^E)$ if $M = 1$.

6 Results and Comparative Statics

The comparative statics have two components. First, the MAH indicator adds the entrusted-production route. Second, conditional on the route being available, the continuous friction τ_i^E affects the value of that route. The derivative-based results are with respect to τ_i^E .

Lemma 1 (Route-set expansion). *For every firm i and every finite τ_i^E ,*

$$\Gamma_i^1(k_i, \tau_i^E) \geq \Gamma_i^0(k_i). \quad (36)$$

Proof. The pre-MAH value in equation (32) maximizes over $\{I, T, A\}$. The post-MAH value in equation (31) maximizes over the larger set $\{I, E, T, A\}$. The maximum over a superset cannot be lower. $\square$

Lemma 2 (Derivative of opportunity value with respect to entrusted-production friction). *Conditional on the entrusted route being feasible, the opportunity value is weakly decreasing in τ_i^E :*

$$\frac{\partial \Gamma_i^1(k_i, \tau_i^E)}{\partial \tau_i^E} \leq 0. \quad (37)$$

If entrusted production is the unique value-maximizing route, then

$$\frac{\partial \Gamma_i^1(k_i, \tau_i^E)}{\partial \tau_i^E} = -1. \quad (38)$$

If another route is strictly value-maximizing, then

$$\frac{\partial \Gamma_i^1(k_i, \tau_i^E)}{\partial \tau_i^E} = 0. \quad (39)$$

At ties, the subgradient lies in $[-1, 0]$.

Proof. From equation (31), only the entrusted-production term

$$R - C^E - \tau_i^E + v \quad (40)$$

depends on τ_i^E . The envelope theorem for the maximum operator gives equations (38) and (39). The subgradient statement follows from convex analysis of a pointwise maximum. $\square$

Proposition 1 (Firm-level R&D response). *Conditional on MAH route availability, optimal R&D intensity is weakly decreasing in the entrusted-production friction:*

$$\frac{\partial x_i^*}{\partial \tau_i^E} = \frac{\beta a_i}{\kappa} \frac{\partial \Gamma_i^1(k_i, \tau_i^E)}{\partial \tau_i^E} \leq 0. \quad (41)$$

Therefore a reduction in τ_i^E weakly raises x_i^* .

Proof. Differentiate equation (33) with respect to τ_i^E and apply Lemma 2. If entrusted production is uniquely optimal, $\partial x_i^* / \partial \tau_i^E = -\beta a_i / \kappa < 0$. $\square$

Proposition 2 (Original-drug count response). *The expected number of successful original-drug opportunities is weakly decreasing in the entrusted-production friction:*

$$\frac{\partial \lambda_i^*}{\partial \tau_i^E} = \frac{\beta a_i^2}{\kappa} \frac{\partial \Gamma_i^1(k_i, \tau_i^E)}{\partial \tau_i^E} \leq 0. \quad (42)$$

Therefore a reduction in τ_i^E weakly raises expected original-drug opportunities or certifies.

Proof. Differentiate equation (34) with respect to τ_i^E and apply Lemma 2. $\square$

Proposition 3 (Binary MAH effect with a friction function). *Let the post-MAH entrusted-production friction be*

$$\tau_i^E(M) = \bar{\tau}_i^E - \Delta \tau_i M, \quad \Delta \tau_i \geq 0. \quad (43)$$

The total MAH effect on firm i 's opportunity value can be written as

$$\Delta \Gamma_i^{MAH} = \underbrace{\Gamma_i^1(k_i, \bar{\tau}_i^E) - \Gamma_i^0(k_i)}_{\text{route-set expansion}} + \underbrace{\left[\Gamma_i^1(k_i, \bar{\tau}_i^E - \Delta \tau_i) - \Gamma_i^1(k_i, \bar{\tau}_i^E) \right]}_{\text{friction reduction}} \geq 0. \quad (44)$$

The first term is weakly nonnegative by Lemma 1. The second term is weakly nonnegative by Lemma 2.

Proof. Add and subtract $\Gamma_i^1(k_i, \bar{\tau}_i^E)$ from $\Gamma_i^1(k_i, \bar{\tau}_i^E - \Delta\tau_i) - \Gamma_i^0(k_i)$. Lemma 1 gives non-negativity of the route-set term. Since $\bar{\tau}_i^E - \Delta\tau_i \leq \bar{\tau}_i^E$, Lemma 2 gives nonnegativity of the friction-reduction term. $\square$

7 Incumbents, Entrants, and Aggregate Drug Counts

Let $z = (a, k, S, \tau^T)$ collect fixed firm characteristics. Let $H_I(z, n)$ be the distribution of incumbents and $G_N(z)$ be the distribution of potential entrants. Let f^e be an entry cost with conditional cumulative distribution $F_e(\cdot | z)$ and density $f_e(\cdot | z)$.

The stationary entrant value, excluding old stock, is

$$A(z, M) = \frac{\beta^2 a^2}{2\kappa(1 - \beta)} \left[\Gamma^M(k) \right]^2. \quad (45)$$

A potential entrant enters if $A(z, M) \geq f^e$. Therefore the mass of entrants is

$$N_E(M) = \int F_e(A(z, M) | z) dG_N(z). \quad (46)$$

Conditional on MAH route availability, differentiating with respect to τ^E gives

$$\frac{\partial A(z, 1)}{\partial \tau^E} = \frac{\beta^2 a^2}{\kappa(1 - \beta)} \Gamma^1(k, \tau^E) \frac{\partial \Gamma^1(k, \tau^E)}{\partial \tau^E} \leq 0. \quad (47)$$

Thus

$$\frac{\partial N_E(1)}{\partial \tau^E} = \int f_e(A(z, 1) | z) \frac{\partial A(z, 1)}{\partial \tau^E} dG_N(z) \leq 0. \quad (48)$$

This is the standard entry logic that higher post-entry value attracts additional entrants (Bresnahan and Reiss, 1991; Berry, 1992), here applied to research-oriented entry.

Aggregate expected original-drug opportunities from incumbents are

$$\Lambda_I(M) = \int \frac{\beta a^2}{\kappa} \Gamma^M(k) dH_I(z, n). \quad (49)$$

Aggregate expected opportunities from entrants are

$$\Lambda_N(M) = \int F_e(A(z, M) | z) \frac{\beta a^2}{\kappa} \Gamma^M(k) dG_N(z). \quad (50)$$

Total aggregate expected original-drug opportunities are

$$\Lambda^{agg}(M) = \Lambda_I(M) + \Lambda_N(M). \quad (51)$$

Conditional on MAH route availability, the incumbent-margin derivative is

$$\frac{\partial \Lambda_I(1)}{\partial \tau^E} = \int \frac{\beta a^2}{\kappa} \frac{\partial \Gamma^1(k, \tau^E)}{\partial \tau^E} dH_I(z, n) \leq 0. \quad (52)$$

The entrant-margin derivative is

$$\begin{aligned} \frac{\partial \Lambda_N(1)}{\partial \tau^E} &= \int F_e(A(z, 1) | z) \frac{\beta a^2}{\kappa} \frac{\partial \Gamma^1(k, \tau^E)}{\partial \tau^E} dG_N(z) \\ &\quad + \int \frac{\beta a^2}{\kappa} \Gamma^1(k, \tau^E) f_e(A(z, 1) | z) \frac{\partial A(z, 1)}{\partial \tau^E} dG_N(z) \leq 0. \end{aligned} \quad (53)$$

Both terms are weakly negative by Lemma 2 and equation (47). Therefore,

$$\frac{\partial \Lambda^{agg}(1)}{\partial \tau^E} \leq 0. \quad (54)$$

A reduction in entrusted-production friction weakly increases aggregate original-drug counts through both incumbent intensity and entrant entry.

Proposition 4 (Aggregate MAH effect). *For any joint distribution of research ability and manufacturing ability, and for any conditional entry-cost distribution, MAH weakly raises aggregate expected original-drug opportunities through route-set expansion and friction reduction. Conditional on the entrusted route being feasible,*

$$\frac{\partial \Lambda^{agg}(1)}{\partial \tau^E} \leq 0, \quad (55)$$

so $d\tau^E < 0$ implies $d\Lambda^{agg} \geq 0$. The increase is strict if a positive-measure set of incumbents or potential entrants has a > 0 , positive operation or entry value, and the entrusted route is relevant in the route-choice maximum.

Proof. Equations (52) and (53) imply equation (54). Route-set expansion is weakly positive by Lemma 1. Strictness follows when $\partial \Gamma^1 / \partial \tau^E < 0$ on a positive-measure set or when adding the entrusted route strictly raises the maximum for a positive-measure set. $\square$

8 Calibration Strategy

The model is not intended to identify every primitive. A feasible calibration should discipline composite objects: the decline in the entrusted-production wedge, the scale of R&D arrival productivity, and the entry-cost distribution. This is a transparent mechanism calibration rather than full structural estimation.

First, route-use data can discipline the wedge decline. The deterministic route choice can be smoothed only for calibration using a logit inclusive value (McFadden, 1974; Berry,

1994). Conditional on comparing entrusted production and internal production among retained products,

$$\log \left(\frac{s_t^E}{1 - s_t^E} \right) = \frac{G_t^E - G_t^I}{\sigma_r} = \frac{C^I(k) - C^E - \tau_t^E}{\sigma_r}, \quad (56)$$

where s_t^E is the observed holder-producer split share among comparable original-drug products and σ_r is a route-choice scale. If $\Delta\tau = \tau_{pre}^E - \tau_{post}^E > 0$, then

$$\frac{\Delta\tau}{\sigma_r} = \log \left(\frac{s_{post}^E}{1 - s_{post}^E} \right) - \log \left(\frac{s_{pre}^E}{1 - s_{pre}^E} \right). \quad (57)$$

This moment identifies the wedge decline relative to the route-choice scale. It does not separately identify legal, search, quality, and contracting components. If the pre-MAH MAH-style entrusted route is structurally unavailable and the corresponding split share is zero, the log-odds expression should be implemented with a small pseudo-count or replaced by post-MAH variation in exposure or implementation intensity. The theoretical comparative statics do not require this logit smoothing; it is only a calibration device.

Second, baseline original-drug counts discipline the composite R&D arrival scale. From equation (34),

$$\bar{\lambda}_0^O = \mathbb{E} \left[\frac{\beta a^2}{\kappa} \Gamma^0(k) \right] \quad (58)$$

for pre-MAH counts. Without project-level R&D cost data, the model cannot separately identify a , κ , and Γ . A practical calibration normalizes one object, such as $\kappa = 1$ or $\Gamma^0(k) = 1$ for a benchmark firm, and interprets $\beta a^2 / \kappa$ as a composite innovation-arrival scale.

Third, entrant counts discipline the entry-cost distribution. From equation (46),

$$N_E(M) = \int F_e(A(z, M) \mid z) dG_N(z). \quad (59)$$

A simple one-parameter distribution for F_e can be chosen to match the pre-MAH entry rate and then used to evaluate whether the post-MAH entry response is consistent with the estimated wedge decline. This step should be reported as sensitivity analysis rather than full structural identification.

The calibration can claim that observed route-use, original-drug counts, and research-oriented entry are consistent with a decline in the entrusted-production wedge. It cannot claim to separately estimate contract-manufacturing prices, legal compliance costs, or firm-specific project success probabilities without additional data.

9 Possible Data Sources and Model Objects

The minimum product-level data are drug registration and approval records with product name, approval date, registration category, original-drug classification, holder, producer or manufacturer, dosage form, and indication or therapeutic class. These records are needed to construct original-drug counts and holder-producer split. Relevant official sources include NMPA and CDE registration and approval data, with careful cleaning of holder and producer names.

The minimum firm-level data are legal-entity identifiers, founding date, location, ownership, business scope, observed first drug project, and proxies for research and manufacturing ability. Research ability a_i can be proxied by pre-reform original-drug applications, patents, clinical trial sponsorship, or R&D expenditure where available. Manufacturing ability k_i can be proxied by Drug Manufacturing Certificates, GMP or manufacturing-site information, dosage-form scope, production license records, or manufacturing assets. Firm registries such as the National Enterprise Credit Information Publicity System and commercial databases such as Tianyancha or Qichacha can help standardize firm identities.

The route-use outcome should identify whether the MAH or holder differs from the actual producer. The main proxy is

$$Split_p = \mathbf{1}\{Holder_p \neq Producer_p\}. \quad (60)$$

This proxy is not the policy treatment. It is an observed outcome that maps to the entrusted-production route for realized products.

Useful additional data include CDE clinical trial or IND/CTA records, CNIPA patent data, NRDL/reimbursement status, centralized procurement data, hospital procurement or sales databases where accessible, and regional policy exposure measures. These data help separate the MAH commercialization channel from broader market-size or demand-side changes.

10 Conclusion

The model gives a simple mechanism. MAH adds the regulated route through which an innovating holder can retain marketing authorization while entrusting production to a qualified manufacturer. Conditional on the route being available, a lower entrusted-production friction raises the route-option value Γ for firms for which the entrusted route is relevant. Because expected next-period route value enters the Bellman, current R&D intensity rises. Since the expected number of new original-drug opportunities is produced by $\lambda = ax$, higher R&D intensity raises expected original-drug counts. Aggregating across

incumbents and entrants gives an aggregate innovation response. The model does not require project-level R&D investment, observed project-specific success probabilities, or a full equilibrium model of contract manufacturing.

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11 Appendix

A Overview of Calibration Strategy

This appendix explains the logic behind the calibration strategy in Section 8. The objective is not to turn the theory section into a full dynamic structural estimation exercise. The objective is narrower: to use a small set of empirical moments that are close to the mechanism to discipline a small set of composite model objects.

The calibration targets three objects:

- (i) the decline in the entrusted-production wedge;
- (ii) the composite scale that converts R&D intensity into original-drug opportunities;
- (iii) the distribution of research-oriented entry costs.

Therefore, Section 8 should be read as a **mechanism calibration**, not as full structural identification. It asks whether the observed route-use patterns, original-drug counts, and research-oriented entry patterns are consistent with the mechanism that MAH lowered the governance and compliance friction of the external commercialization route. It does not claim to separately identify legal compliance costs, search costs, quality-management costs, contract-manufacturing prices, or firm-specific project success probabilities.

B Definition of Moments

A moment is a statistic that can be computed both in the data and in the model. For example, in the data one may compute

$$s_{post}^{E,data} = \frac{\#\{\text{post-reform original-drug records with different holder and producer}\}}{\#\{\text{comparable post-reform original-drug records}\}}.$$

Given parameters, the model can generate the corresponding object:

$$s_{post}^{E,model}(\theta).$$

The calibration logic is to choose parameters θ such that

$$m^{model}(\theta) \approx m^{data}.$$

The moment m can be a route-use share, an original-drug count, an entrant count, an original-drug share, or a holder–producer split. A moment is not a new structural variable. It is the empirical bridge between the model and the data.

C Calibration Blocks

The three calibration blocks in Section 8 are summarized in Table 1.

These blocks correspond to the same mechanism chain:

$$\tau^E \Downarrow \Rightarrow \Gamma \Uparrow \Rightarrow x^* \Uparrow \Rightarrow \lambda^* \Uparrow \Rightarrow \Lambda^{agg} \Uparrow \Rightarrow N_E \Uparrow.$$

Here τ^E is the governance and compliance friction of the external commercialization route, Γ is the expected commercialization value of a successful original-drug opportunity,

Table 1: Three Calibration Blocks

Calibration block	Data moment	Model moment	Disciplined object
Route-use data	Holder–producer split share; entrusted-production route-use share	$\log \frac{s_t^E}{1-s_t^E} = \frac{G_t^E - G_t^I}{\sigma_r}$	$\Delta\tau^E/\sigma_r$
Original-drug counts	Baseline original-drug approvals, realized products, or applications	$\bar{\lambda}_0^O = \mathbb{E}\left[\frac{\beta a^2}{\kappa}\Gamma^0(k)\right]$	Composite innovation-arrival scale $\beta a^2/\kappa$
Entrant counts	Research-oriented entrant counts or proxies	$N_E(M) = \int F_e(A)dG_N$	Entry-cost distribution F_e

x^* is optimal original-drug R&D intensity, λ^* is the arrival rate of original-drug opportunities, Λ^{agg} is aggregate expected original-drug opportunities, and N_E is research-oriented entry.

D Route-Use Moment and Entrusted-Production Wedge

D.1 The Data Moment

The central data object for route use is the holder–producer split. For product record p , define

$$Split_p = \mathbb{I}\{Holder_p \neq Producer_p\}.$$

Restricting attention to original-drug records, define

$$s_t^{E,data} = \frac{\sum_{p \in t} Split_p O_p}{\sum_{p \in t} O_p},$$

where $O_p = 1$ if product p is classified as an original drug or satisfies the paper’s baseline original-drug proxy. This share is not the policy treatment. It is an observed outcome that proxies the use of an external commercialization route while retaining authorization.

D.2 Why Route Share Maps to τ^E

After the reform, when the entrusted route is feasible, a firm compares commercialization routes after a successful original-drug opportunity is realized. To isolate the logic, consider only internal production and entrusted production among retained products:

$$G_t^I = R_t - C^I(k) + v,$$

$$G_t^E = R_t - C^E - \tau_t^E + v.$$

Subtracting internal production from entrusted production gives

$$G_t^E - G_t^I = C^I(k) - C^E - \tau_t^E.$$

Three points are important. First, R_t cancels because both routes commercialize the drug. Second, v cancels because both routes allow the innovator to retain the original-drug asset. Third, the remaining difference is internal production cost, direct entrusted-production cost, and the entrusted-route friction. Thus, a lower τ_t^E makes entrusted production more attractive relative to internal production.

For calibration only, suppose this deterministic route choice is smoothed by a binary logit:

$$s_t^E = \frac{\exp(G_t^E/\sigma_r)}{\exp(G_t^E/\sigma_r) + \exp(G_t^I/\sigma_r)}.$$

Then the odds ratio is

$$\frac{s_t^E}{1 - s_t^E} = \exp\left(\frac{G_t^E - G_t^I}{\sigma_r}\right),$$

and taking logs yields

$$\log\left(\frac{s_t^E}{1 - s_t^E}\right) = \frac{G_t^E - G_t^I}{\sigma_r} = \frac{C^I(k) - C^E - \tau_t^E}{\sigma_r}.$$

If $C^I(k)$ and C^E are stable across the comparison, or are handled through comparable samples and controls, the pre-post difference in log-odds is

$$\frac{\Delta\tau}{\sigma_r} = \log\left(\frac{s_{post}^E}{1 - s_{post}^E}\right) - \log\left(\frac{s_{pre}^E}{1 - s_{pre}^E}\right),$$

where

$$\Delta\tau = \tau_{pre}^E - \tau_{post}^E > 0.$$

This equation does not say that the route-use share equals τ^E . It says that the route-use share, through the route-choice equation, disciplines the change in the relative attractiveness of the entrusted route. More precisely, it identifies

$$\Delta\tau^E/\sigma_r,$$

namely the decline in the external-route wedge relative to the route-choice scale.

D.3 What This Block Can and Cannot Identify

This block can support the following statement: if the holder-producer split among original-drug records rises after MAH, and this rise cannot be explained by changes

in product composition, therapeutic class, region, or firm capability composition, the pattern is consistent with a decline in τ^E .

It cannot support the following stronger statement: the data separately identify legal compliance costs, search costs, quality-agreement costs, contracting costs, and contract-manufacturing prices. These components enter the route-choice payoff as a composite wedge. Without additional data on quality agreements, auditing costs, contract terms, manufacturing prices, and firm-level regulatory execution, they cannot be separately recovered.

D.4 What If Pre-Reform Route-Use Data Are Unavailable?

This is the most important practical issue. If the MAH-style entrusted route was institutionally unavailable before the reform, then

$$s_{pre}^E = 0.$$

In that case,

$$\log\left(\frac{s_{pre}^E}{1 - s_{pre}^E}\right) = \log(0)$$

is undefined. The formula cannot be used mechanically. There are three disciplined responses.

Response 1: Treat the pre-reform regime as a route-set-expansion boundary case. Before MAH,

$$\mathcal{R}(0) = \{I, T, A\}.$$

After MAH,

$$\mathcal{R}(1) = \{I, E, T, A\}.$$

Thus there is no comparable pre-reform entrusted-route share. The baseline interpretation should emphasize that MAH first expands the feasible route set and then, conditional on route feasibility, lowers the governance and compliance friction of the route.

Response 2: Use a small pseudo-count only as a computational device. For example, use

$$\tilde{s}_{pre}^E = \frac{0.5}{N_{pre} + 1}$$

instead of zero. This makes the log-odds expression computable, but the pseudo-share should not be interpreted as a real pre-reform route-use rate. It should be reported only as a sensitivity check.

Response 3: Use post-reform exposure or implementation-intensity variation. If the route is unavailable before MAH, one can use post-MAH variation across regions,

dosage forms, therapeutic classes, or firm types. For example, regions or dosage-form segments with greater pre-reform qualified manufacturing capacity may experience a larger effective decline in external commercialization friction. This variation can discipline the effective change in τ^E without imposing a nonexistent pre-reform route share.

E Original-Drug Counts and R&D Arrival Scale

E.1 The Model Equation

The firm chooses original-drug R&D intensity x_i . The arrival rate of original-drug opportunities is

$$\lambda_i(x_i) = a_i x_i.$$

Given the expected commercialization value Γ_i^M of a successful opportunity, the first-order condition gives

$$x_i^*(M) = \frac{\beta a_i}{\kappa} \Gamma_i^M.$$

Therefore, the optimal arrival rate is

$$\lambda_i^*(M) = a_i x_i^*(M) = \frac{\beta a_i^2}{\kappa} \Gamma_i^M.$$

Taking expectations in the pre-reform or baseline regime gives

$$\bar{\lambda}_0^O = \mathbb{E} \left[\frac{\beta a_i^2}{\kappa} \Gamma^0(k) \right].$$

E.2 The Data Moment

The most direct moment is the baseline number of original-drug outcomes:

$$\bar{\lambda}_0^{O,data} = \frac{\#\{\text{pre-reform original-drug records}\}}{\#\{\text{pre-reform years}\}}.$$

At the firm-year level, define

$$Y_{it}^{O,base} = \sum_{p \in (i,t)} O_p^{base}.$$

If the current dataset is an approval-side realized-product file, this object should be interpreted as realized original-drug products, not latent idea arrival. If future data include CDE applications, IND/CTA records, or clinical-trial registrations, those objects can be used to construct moments closer to upstream innovation-opportunity arrival.

E.3 Why This Moment Only Disciplines a Composite Parameter

From

$$\lambda_i^* = \frac{\beta a_i^2}{\kappa} \Gamma_i,$$

original-drug counts depend on three objects:

- (i) a_i : research ability;
- (ii) κ : the marginal cost scale of R&D effort;
- (iii) Γ_i : the commercialization value of a successful opportunity.

If the data only reveal realized original-drug counts, the model cannot separately identify a_i , κ , and Γ_i . High original-drug counts may reflect strong scientific capability, low R&D effort costs, or high commercialization value. Without project-level R&D costs, project success rates, project values, or rich firm-capability data, these objects cannot be separately recovered.

The disciplined approach is to normalize one object. For example,

$$\kappa = 1,$$

or for a benchmark firm,

$$\Gamma^0(k) = 1.$$

Then

$$\frac{\beta a^2}{\kappa}$$

is interpreted as a composite innovation-arrival scale.

E.4 What If Pre-Reform Original-Drug Counts Are Sparse?

Sparse pre-reform approvals do not invalidate the mechanism. Drug development cycles are long, and final approvals are downstream outcomes. Pre-reform approval counts can therefore be noisy low-frequency moments. Disciplined responses include:

1. using multi-year averages rather than year-by-year moments;
2. using IND/CTA records, clinical-trial registrations, CDE acceptance numbers, or original-drug applications as upstream proxies;
3. separating realized approvals from upstream applications as distinct moments;
4. reporting robustness under alternative original-drug definitions, such as broad new-drug definitions, strict class-I innovative drugs, or inclusion/exclusion of improved new drugs.

The writing boundary is simple: approval-side data calibrate realized original-drug product counts; application-side data are closer to innovation-opportunity arrival.

F Entry Counts and Entry-Cost Distribution

F.1 The Model Equation

A potential entrant enters if post-entry value exceeds entry cost:

$$A(z, M) \geq f_e.$$

The post-entry value is

$$A(z, M) = \frac{\beta^2 a^2}{2\kappa(1 - \beta)} [\Gamma^M(k)]^2.$$

If f_e has conditional distribution function $F_e(\cdot | z)$, the entry probability is

$$\Pr(f_e \leq A(z, M) | z) = F_e(A(z, M) | z).$$

Aggregating over potential entrants gives

$$N_E(M) = \int F_e(A(z, M) | z) dG_N(z).$$

F.2 The Data Moment

The ideal data moment is the number of research-oriented entrants, not the number of all newly registered pharmaceutical firms. Examples include:

- firms that appear for the first time in original-drug applications;
- firms that sponsor clinical trials for the first time;
- firms that first appear as MAH/holder for original-drug records;
- firms without traditional production capacity that enter original-drug R&D or commercialization records.

Define

$$Entry_{it}^R = 1$$

if firm i first enters observable original-drug R&D or original-drug realization activity in year t . Then the data moment is

$$N_{E,t}^{data} = \sum_i Entry_{it}^R.$$

F.3 Why This Step Should Be Sensitivity or a Future Module

A current approval-side realized-product file is typically not a firm-entry panel. Observing that a firm first appears in an approval record is not equivalent to observing true firm entry or research-oriented entry. There are three reasons:

1. the firm may have been founded long before its first approval;
2. the firm may have upstream R&D activity that has not yet reached final approval;
3. name standardization, acquisitions, subsidiaries, and holder-entity changes can affect the first observed year.

Therefore, entrant-count calibration should not be a baseline result in the current data environment. It should be kept as a future data module or as a sensitivity appendix. A serious calibration of F_e requires a merged firm-year entry panel, business registration data, CDE application data, and firm-capability data.

F.4 How to Calibrate F_e

One can assume a simple one-parameter distribution. For example,

$$f_e \sim \text{Exponential}(\mu),$$

so that

$$F_e(A) = 1 - \exp(-A/\mu).$$

The pre-MAH entry rate can be used to match μ , and the post-MAH change in Γ^M can then be used to predict the post-MAH entry response. A lognormal distribution is also possible, but if the data are thin, it is better not to introduce too many shape parameters.

The interpretation should be modest: this step does not estimate each potential entrant's true entry cost. It checks whether the entry margin moves in the direction predicted by a lower external commercialization wedge.

G Handling Missing Pre-Reform Data

Missing or weak pre-reform data are not a minor nuisance. They are central to the calibration design. The empirical architecture should therefore be modular.

G.1 Module A: Current Approval-Side Realized-Product Moments

The most credible baseline moments are:

1. realized original-drug product counts;

2. original-drug share among realized products;
3. holder–producer split;
4. split-based route-use proxy among original-drug records.

These objects can be constructed from the current approval-side matching file, but they are realized-product objects rather than full R&D-process objects.

G.2 Module B: Future Application and Status Panel

Once CDE applications, IND/CTA records, clinical trials, withdrawn projects, pending projects, rejected projects, and approved projects are merged, one can construct:

1. application-to-approval conversion;
2. project realization probabilities;
3. upstream original-drug opportunity arrival;
4. delay and conversion moments.

These moments are closer to λ_i and commercialization success.

G.3 Module C: Future Firm-Capability and Entry Panel

Once business-registration records, production licenses, GMP or dosage-form scope, R&D expenditure, patents, and clinical-trial sponsorship are merged, one can construct:

1. pre-reform research capability;
2. pre-reform manufacturing capability;
3. research-oriented entry;
4. capability-bin-specific route-use and realization moments.

These moments are needed to calibrate z , F_e , and heterogeneous responses more credibly.

H Recommended Scope for Appendix

The following scope limits should be explicit after Section 8:

1. **What can be claimed:** changes in route use, original-drug counts, and research-oriented entry are consistent with a decline in the entrusted-production wedge.
2. **What cannot be claimed:** the current data do not separately estimate contract-manufacturing prices, legal compliance costs, firm-specific project success probabilities, or firm-specific true governance costs.

3. **Current baseline:** use approval-side realized-product moments for mechanism calibration.
4. **Future upgrade:** once application panels, firm-year entry panels, and capability panels are merged, the exercise can be expanded toward a fuller SMM or indirect-inference design.

I Suggested Text for Main Paper

The following paragraph can be used directly in the appendix or adapted for the main text:

The calibration strategy in Section 8 should be interpreted as mechanism calibration rather than full structural estimation. The route-use moment disciplines the decline in the external entrusted-production wedge relative to the route-choice scale; the baseline original-drug count moment disciplines the composite scale that converts R&D effort into original-drug opportunities; and the research-oriented entry moment, when available, disciplines the entry-cost distribution. Because the MAH-style entrusted route may be structurally unavailable before the reform, the pre-reform split share should not be mechanically inserted into a log-odds formula. The baseline interpretation treats the reform as route-set expansion plus conditional friction reduction, while pseudo-counts or post-MAH exposure variation are used only as robustness devices. The current approval-side data can support realized original-drug counts and holder–producer split, but they do not directly identify latent idea arrival, project-level success probabilities, or true research-oriented entry. The calibration exercise is therefore a disciplined mechanism calibration, not full structural identification.

J Appendix Conclusion

The correct interpretation of Section 8 is that data moments discipline composite model objects rather than every structural primitive. Route-use data discipline $\Delta\tau^E/\sigma_r$, original-drug counts discipline $\beta a^2/\kappa$, and research-oriented entry disciplines F_e . When many pre-reform data objects are unavailable, the baseline must remain modest: the current exercise is approval-side mechanism calibration, while richer entry and state calibration should be reserved for future merged data modules.